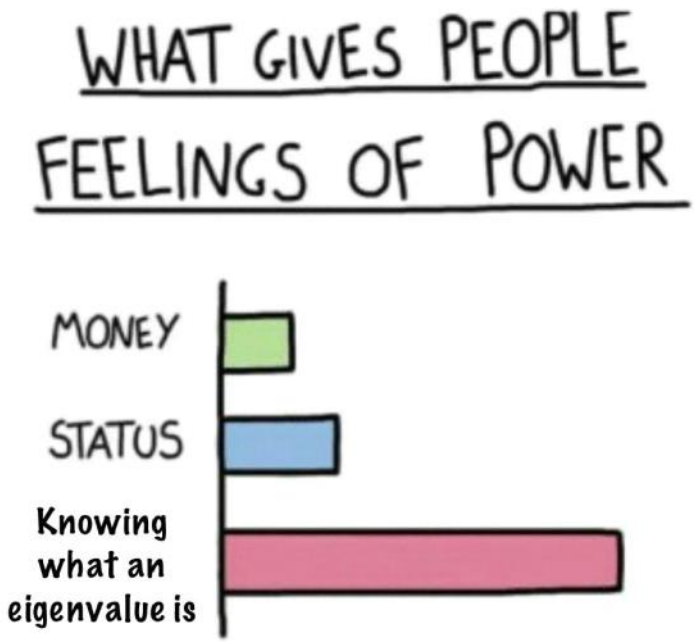
Eigenvalue & Eigenvector

Monday, 9 February 2026 12:27 pm



5.1.1 Eigenvalues and Eigenvectors

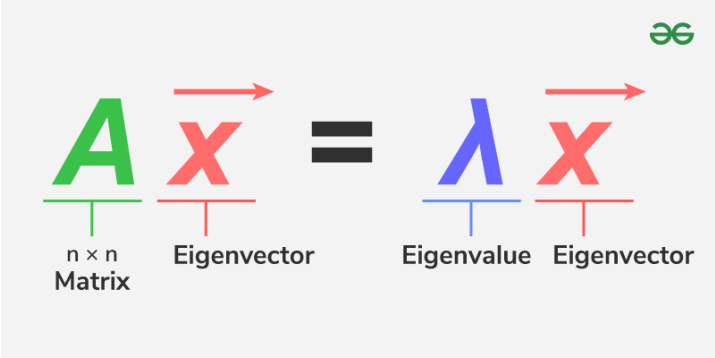
Here is the most important definition in this text.

Essential Definition. Let A be an $n \times n$ matrix.

1. An **eigenvector** of A is a *nonzero* vector v in $\mathbf{R}^n$ such that $Av = \lambda v$, for some scalar λ.

2. An **eigenvalue** of A is a scalar λ such that the equation $Av = \lambda v$ has a *non-trivial* solution.

If $Av = \lambda v$ for $v \neq 0$, we say that λ is the **eigenvalue for** v, and that v is an **eigenvector for** λ.



- **Data Science and Machine Learning (Dimensionality Reduction):** Principal Component Analysis (PCA) uses eigenvectors of covariance matrices to identify key features in high-dimensional data, reducing complexity while preserving information.
- **Structural Engineering (Vibration Analysis):** Eigenvalues determine the "natural frequencies" of structures (bridges, buildings), while eigenvectors represent the corresponding "mode shapes," helping engineers design for stability and avoid resonance.
- **Computer Science (PageRank Algorithm):** Google's original search algorithm computes the eigenvector corresponding to the largest eigenvalue of the web link matrix to rank web pages by importance.
- **Computer Vision (Facial Recognition):** "Eigenfaces" use eigenvectors to represent facial features, allowing for efficient facial recognition and data compression.
- **Control Systems and Stability:** In engineering, eigenvalues of a system matrix determine whether a system (like a drone or chemical reactor) is stable, unstable, or oscillating.
- **Quantum Mechanics:** Physical observables are represented by operators, and their eigenvalues correspond to the possible measured values (like energy levels), while eigenvectors represent the corresponding quantum states.
- **Principal Component Analysis (PCA):** Eigenvectors are crucial for finding the directions of maximum variance in a dataset, which is used to compress data and

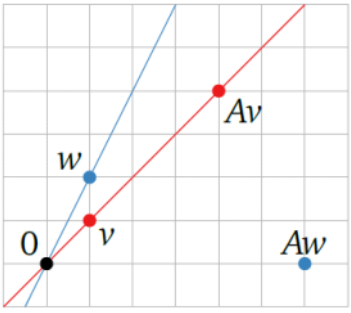
Eigenvectors are *by definition nonzero*. Eigenvalues may be equal to zero.

We do not consider the zero vector to be an eigenvector: since $A0 = 0 = \lambda 0$ for *every* scalar λ, the associated eigenvalue would be undefined.

Example (Verifying eigenvectors). Consider the matrix

$$A = \begin{pmatrix} 2 & 2 \\ -4 & 8 \end{pmatrix} \quad \text{and vectors} \quad v = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad w = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

Which are eigenvectors? What are their eigenvalues?



v is an eigenvector

w is not an eigenvector

Example (Verifying eigenvectors). Consider the matrix

$$A = \begin{pmatrix} 0 & 6 & 8 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \end{pmatrix} \quad \text{and vectors} \quad v = \begin{pmatrix} 16 \\ 4 \\ 1 \end{pmatrix} \qquad w = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$$

Which are eigenvectors? What are their eigenvalues?

Example (An eigenvector with eigenvalue 0). Let

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix} \qquad v = \begin{pmatrix} -3 \\ 1 \end{pmatrix}.$$

Is v an eigenvector of A? If so, what is its eigenvalue?

Eigenspaces

$$\begin{aligned} & Av = \lambda v \\ \iff\ & Av - \lambda v = 0 \\ \iff\ & Av - \lambda I_n v = 0 \\ \iff\ & (A - \lambda I_n)v = 0. \end{aligned}$$

$$A = \begin{pmatrix} 7 & 1 & 3 \\ -3 & 2 & -3 \\ -3 & -2 & -1 \end{pmatrix},$$

$$\begin{pmatrix} 7 & 1 & 3 \\ -3 & 2 & -3 \\ -3 & -2 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

$$\begin{cases} 7x + y + 3z = \lambda x \\ -3x + 2y - 3z = \lambda y \\ -3x - 2y - z = \lambda z \end{cases} \longrightarrow \begin{cases} (7-\lambda)x + y + 3z = 0 \\ -3x + (2-\lambda)y - 3z = 0 \\ -3x - 2y + (-1-\lambda)z = 0. \end{cases}$$

This is the same as the homogeneous matrix equation

$$\begin{pmatrix} 7-\lambda & 1 & 3 \\ -3 & 2-\lambda & -3 \\ -3 & -2 & -1-\lambda \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0,$$

i.e., $(A - \lambda I_3)v = 0$.

Definition. Let A be an $n \times n$ matrix, and let λ be an eigenvalue of A. The **λ-eigenspace** of A is the solution set of $(A-\lambda I_n)v = 0$, i.e., the subspace $\mathrm{Nul}(A-\lambda I_n)$.

The λ-eigenspace is a subspace because it is the null space of a matrix, namely, the matrix $A-\lambda I_n$. This subspace consists of the zero vector and all eigenvectors of A with eigenvalue λ.

Note. Since a nonzero subspace is infinite, *every eigenvalue has infinitely many eigenvectors*. (For example, multiplying an eigenvector by a nonzero scalar gives another eigenvector.) On the other hand, there can be at most n linearly independent eigenvectors of an $n \times n$ matrix, since $\mathbf{R}^n$ has dimension n.

Example (Computing eigenspaces)**.** For each of the numbers $\lambda = -2, 1, 3$, decide if λ is an eigenvalue of the matrix

$$A = \begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix},$$

Solution. The number 3 is an eigenvalue of A if and only if $\mathrm{Nul}(A-3I_2)$ is nonzero. Hence, we have to solve the matrix equation $(A-3I_2)v = 0$. We have

$$A - 3I_2 = \begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix} - 3\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & -4 \\ -1 & -4 \end{pmatrix}.$$

The reduced row echelon form of this matrix is

$$\begin{pmatrix} 1 & 4 \\ 0 & 0 \end{pmatrix} \xrightarrow[\text{form}]{\text{parametric}} \begin{cases} x = -4y \\ y = y \end{cases} \xrightarrow[\text{vector form}]{\text{parametric}} \begin{pmatrix} x \\ y \end{pmatrix} = y\begin{pmatrix} -4 \\ 1 \end{pmatrix}.$$

Since y is a free variable, the null space of $A-3I_2$ is nonzero, so 3 is an eigenvector. A basis for the 3-eigenspace is $\left\{\left(\begin{smallmatrix} -4 \\ 1 \end{smallmatrix}\right)\right\}$.

$$\begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix}\begin{pmatrix} -4 \\ 1 \end{pmatrix} = \begin{pmatrix} -12 \\ 3 \end{pmatrix} = 3\begin{pmatrix} -4 \\ 1 \end{pmatrix}.$$

The number 1 is an eigenvalue of A if and only if $\mathrm{Nul}(A-I_2)$ is nonzero. Hence, we have to solve the matrix equation $(A-I_2)v = 0$. We have

$$A - I_2 = \begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -4 \\ -1 & -2 \end{pmatrix}.$$

This matrix has determinant -6, so it is invertible. By the invertible matrix theorem in Section 3.6, we have $\mathrm{Nul}(A-I_2) = \{0\}$, so 1 is not an eigenvalue.

The eigenvectors of A with eigenvalue -2, if any, are the nonzero solutions of the matrix equation $(A+2I_2)v = 0$. We have

$$A + 2I_2 = \begin{pmatrix} 2 & -4 \\ -1 & -1 \end{pmatrix} + 2\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -4 \\ -1 & 1 \end{pmatrix}.$$

The reduced row echelon form of this matrix is

$$\begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \xrightarrow[\text{form}]{\text{parametric}} \begin{cases} x = y \\ y = y \end{cases} \xrightarrow[\text{vector form}]{\text{parametric}} \begin{pmatrix} x \\ y \end{pmatrix} = y\begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Hence there exist eigenvectors with eigenvalue -2, namely, any nonzero multiple of $\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$. A basis for the -2-eigenspace is $\left\{\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)\right\}$.

Definition. Let A be an $n \times n$ matrix. The **characteristic polynomial** of A is the function $f(\lambda)$ given by

$$f(\lambda) = \det(A - \lambda I_n).$$

Example. Find the characteristic polynomial of the matrix

$$A = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}.$$

Solution. We have

$$\begin{aligned} f(\lambda) = \det(A - \lambda I_2) &= \det\left(\begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}\right) \\ &= \det\begin{pmatrix} 5-\lambda & 2 \\ 2 & 1-\lambda \end{pmatrix} \\ &= (5-\lambda)(1-\lambda) - 2 \cdot 2 = \lambda^2 - 6\lambda + 1. \end{aligned}$$

Example. Find the characteristic polynomial of the matrix

$$A = \begin{pmatrix} 0 & 6 & 8 \\ \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \end{pmatrix}.$$

Solution. We compute the determinant by expanding cofactors along the third column:

$$\begin{aligned} f(\lambda) = \det(A - \lambda I_3) &= \det\begin{pmatrix} -\lambda & 6 & 8 \\ \frac{1}{2} & -\lambda & 0 \\ 0 & \frac{1}{2} & -\lambda \end{pmatrix} \\ &= 8\left(\frac{1}{4} - 0 \cdot -\lambda\right) - \lambda\left(\lambda^2 - 6 \cdot \frac{1}{2}\right) \\ &= -\lambda^3 + 3\lambda + 2. \end{aligned}$$

Example (Finding eigenvalues)**.** Find the eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}.$$

Solution. In the above example we computed the characteristic polynomial of A to be $f(\lambda) = \lambda^2 - 6\lambda + 1$. We can solve the equation $\lambda^2 - 6\lambda + 1 = 0$ using the quadratic formula:

$$\lambda = \frac{6 \pm \sqrt{36-4}}{2} = 3 \pm 2\sqrt{2}.$$

Therefore, the eigenvalues are $3 + 2\sqrt{2}$ and $3 - 2\sqrt{2}$.

To compute the eigenvectors, we solve the homogeneous system of equations $(A - \lambda I_2)x = 0$ for each eigenvalue λ. When $\lambda = 3 + 2\sqrt{2}$, we have

$$\begin{aligned} A - (3+\sqrt{2})I_2 &= \begin{pmatrix} 2-2\sqrt{2} & 2 \\ 2 & -2-2\sqrt{2} \end{pmatrix} \\ &\xrightarrow{R_1 = R_1 \times (2+2\sqrt{2})} \begin{pmatrix} -4 & 4+4\sqrt{2} \\ 2 & -2-2\sqrt{2} \end{pmatrix} \\ &\xrightarrow{R_2 = R_2 + R_1/2} \begin{pmatrix} -4 & 4+4\sqrt{2} \\ 0 & 0 \end{pmatrix} \end{aligned}$$

$$\xrightarrow{R_1 = R_1 \div -4} \begin{pmatrix} 1 & -1-\sqrt{2} \\ 0 & 0 \end{pmatrix}.$$

The parametric form of the general solution is $x = (1+\sqrt{2})y$, so the $(3+2\sqrt{2})$-eigenspace is the line spanned by $\binom{1+\sqrt{2}}{1}$. We compute in the same way that the $(3-2\sqrt{2})$-eigenspace is the line spanned by $\binom{1-\sqrt{2}}{1}$.

$$\mathbf{A} = \begin{bmatrix} -2 & 4 \\ -6 & 8 \end{bmatrix}.$$

$$\mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} -2 & 4 \\ -6 & 8 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} -2-\lambda & 4 \\ -6 & 8-\lambda \end{bmatrix}$$

$$\begin{aligned} p(\lambda) &= \det(\mathbf{A} - \lambda\mathbf{I}) \\ &= \begin{vmatrix} -2-\lambda & 4 \\ -6 & 8-\lambda \end{vmatrix} \\ &= (-2-\lambda)(8-\lambda) + 24 \\ &= \lambda^2 - 6\lambda + 8 \\ &= (\lambda - 4)(\lambda - 2) \end{aligned}$$

Example 22.4. Find the eigenvalues of

$$\mathbf{A} = \begin{bmatrix} -4 & -6 & -7 \\ 3 & 5 & 3 \\ 0 & 0 & 3 \end{bmatrix}.$$

Solution. Compute

$$\mathbf{A} - \lambda\mathbf{I} = \begin{bmatrix} -4 & -6 & -7 \\ 3 & 5 & 3 \\ 0 & 0 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix}$$

$$= \begin{bmatrix} -4-\lambda & -6 & -7 \\ 3 & 5-\lambda & 3 \\ 0 & 0 & 3-\lambda \end{bmatrix}$$

Then

$$\begin{aligned} \det(\mathbf{A} - \lambda\mathbf{I}) &= (-4-\lambda)\begin{vmatrix} 5-\lambda & 3 \\ -\lambda & 3-\lambda \end{vmatrix} - 3\begin{vmatrix} -6 & -7 \\ -\lambda & 3-\lambda \end{vmatrix} \\ &= (-4-\lambda)[(3-\lambda)(5-\lambda) + 3\lambda] - 3[-6(3-\lambda) - 7\lambda] \\ &= \lambda^3 - 4\lambda^2 + \lambda + 6 \end{aligned}$$

Factor the characteristic polynomial:

$$p(\lambda) = \lambda^3 - 4\lambda^2 + \lambda + 6 = (\lambda - 2)(\lambda - 3)(\lambda + 1)$$

Therefore, the eigenvalues of $\mathbf{A}$ are

$$\lambda_1 = 2, \quad \lambda_2 = 3, \quad \lambda_3 = -1.$$

Start with $\lambda_1 = 2$:

$$\mathbf{A} - 2\mathbf{I} = \begin{bmatrix} -6 & -6 & -7 \\ 3 & 3 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$\mathbf{v}_1$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ_1. For $\lambda_2 = 3$:

$$\mathbf{A} - 3\mathbf{I} = \begin{bmatrix} -7 & -6 & -7 \\ 3 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$\mathbf{v}_2$ is an eigenvector of $\mathbf{A}$ with eigenvalue λ_2. Finally, for $\lambda_3 = -1$

$$\mathbf{A} - \lambda_3\mathbf{I} = \begin{bmatrix} -3 & -6 & -7 \\ 3 & 6 & 3 \\ 0 & 0 & 4 \end{bmatrix}$$

space of $\mathbf{A} - \lambda_3\mathbf{I}$ is spanned by

$$\mathbf{v}_3 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Example 22.7. Find the eigenvalues of $\mathbf{A}$ and a basis for each eigenspace.

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 \\ 4 & 2 & 2 \\ -2 & 0 & 1 \end{bmatrix}$$

Does $\mathbb{R}^3$ have a basis of eigenvectors of $\mathbf{A}$?

Solution. The characteristic polynomial of $\mathbf{A}$ is

$$p(\lambda) = \det(\mathbf{A} - \lambda\mathbf{I}) = \lambda^3 - 5\lambda^2 + 8\lambda - 4 = (\lambda - 1)(\lambda - 2)^2$$

and therefore the eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 2$. Notice that although $p(\lambda)$ is a polynomial of degree $n = 3$, it has **only two distinct roots** and hence $\mathbf{A}$ has only two distinct eigenvalues. The eigenvalue $\lambda_2 = 2$ is said to be **repeated** and $\lambda_1 = 1$ is said to be a **simple** eigenvalue. For $\lambda_1 = 1$ one finds that the eigenspace $\text{Null}(\mathbf{A} - \lambda_1\mathbf{I})$ is spanned by

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}$$

and thus $\mathbf{v}_1$ is an eigenvector of $\mathbf{A}$ with eigenvalue $\lambda_1 = 1$. Now consider $\lambda_2 = 2$:

$$\mathbf{A} - 2\mathbf{I} = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 2 \\ -2 & 0 & -1 \end{bmatrix}$$

Row reducing $\mathbf{A} - 2\mathbf{I}$ one obtains

$$\mathbf{A} - 2\mathbf{I} = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 2 \\ -2 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} -2 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore, $\text{rank}(\mathbf{A} - 2\mathbf{I}) = 1$ and thus by the Rank Theorem it follows that $\text{Null}(\mathbf{A} - 2\mathbf{I})$ is a 2-dimensional eigenspace. Performing back substitution, one finds the following basis for the λ_2-eigenspace:

$$\{\mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Therefore, the eigenvectors

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Example 22.9. A 6×6 matrix $\mathbf{A}$ has characteristic polynomial

$$p(\lambda) = \lambda^6 - 4\lambda^5 - 12\lambda^4.$$

Find the eigenvalues of $\mathbf{A}$ and their algebraic multiplicities.

Example 22.11. Find the algebraic and geometric multiplicities of each eigenvalue of the matrix

$$\mathbf{A} = \begin{bmatrix} -7 & 1 & 0 \\ 0 & -7 & 1 \\ 0 & 0 & -7 \end{bmatrix}.$$